PARADIGM JUSTIFICATION STATEMENT

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1. Research Topic and Context

Higher education institutions often identify student dropout only after it occurs, despite holding academic, administrative, temporal, and textual records that may reveal risk earlier. This study will develop an explainable, multimodal, and temporal artificial intelligence framework at an Ecuadorian higher education institution. The artifact will support academic advisors, student welfare units, program leaders, and institutional analysts through timely risk estimates and explanations adapted to their responsibilities. All outputs will support human decisions and access to assistance; they will not trigger automatic or punitive action.

2. Preliminary Research Question

Can an explainable, multimodal, and temporal artificial intelligence framework be designed and evaluated to predict which undergraduate students will no longer be registered at a higher education institution after the four regular academic periods following their first registration, while maintaining predictive performance, fairness, interpretability, and institutional usefulness?

3. Philosophical Position and Chosen Research Approach

The philosophical position is pragmatism because knowledge will be judged by its ability to address a real institutional problem while accounting for empirical performance and practical consequences. Design Science Research is the primary approach because the contribution is an evaluated artifact and transferable design knowledge. The study will follow the rigor, relevance, and evaluation principles of Hevner et al. (2004), the iterative process of Peffers et al. (2007), and the contribution framing of Gregor and Hevner (2013). An embedded single institutional case will provide the context for demonstration, while comparative quantitative evaluation against structured data baselines will provide empirical evidence. Interpretivism is outside the present scope because no lived experience or meaning-making strand is planned. Numeric, temporal, and computationally processed text constitute multimodal analysis rather than mixed methods because the study does not integrate separate quantitative and qualitative inquiries.

4. Implications for Data, Method, Validation, Ethics, and Use

The outcome will be defined before modeling according to the Ecuadorian national accreditation model established by CACES (Consejo de Aseguramiento de la Calidad de la Educación Superior). For each first registration cohort, a student will be classified as a dropout when the student is no longer registered at the institution at the observation point corresponding to the four regular academic periods following first registration. The direct rate follows calendar year reporting. For example, the 2026 indicator evaluates registration continuity during 2025 among students whose first registration occurred in 2023. Classification occurs at the institutional level; therefore, a change of academic program within the same institution will not be considered dropout. The protocol will establish the prediction cutoff before modeling. Every feature must precede that cutoff, and variables that directly encode later nonregistration will be excluded to prevent outcome leakage.

An adapted CRISP-DM and Design Science process will guide problem definition, data preparation, artifact construction, demonstration, evaluation, and refinement. Structured inputs may include admission, enrollment, grades, credits, attendance, financial aid, and learning platform activity. Authorized textual inputs will be limited to records with a clear educational purpose, such as tutoring notes, open-ended survey responses, and student service requests. Access will require approval from the host institution's ethics committee and data governance authority, with informed consent or an approved basis for secondary use. Data will be minimized, deidentified, access controlled, and retained only for the approved period. Risk estimates will remain decision support and will require human review.

Validation will use temporally separated cohorts, appropriate discrimination and calibration measures, and ablations comparing structured only, structured plus temporal, structured plus text, and full multimodal models. Fairness will be assessed through error and calibration differences across lawful and sufficiently represented student groups. Constrained counterfactual explanations based on mutable factors will guide advisors, aggregate SHAP (SHapley Additive exPlanations) summaries will support administrators, and students will receive plain language support options only after human review. Before testing, the institution will define utility criteria for an advisor decision study, including alert actionability, referral appropriateness, explanation clarity, decision time, and acceptable workload. The contribution will comprise the validated framework, its design principles, and responsible use guidance.

5. One Doubt or Tension

Intervention creates a feedback loop because support prompted by the model may prevent the predicted outcome, making later labels partly dependent on earlier predictions. During a pilot, intervention exposure and timing will therefore be logged. Prospective evaluation will use a staged rollout or an untouched holdout cohort where ethically acceptable, and deployment data will not be combined with historical data without accounting for the intervention policy. Missing or uneven textual data, cohort changes, and the single institution context may still limit transferability.

References

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